Exploratory Data Analysis (EDA) Summary

Delinquency Prediction Dataset

1. Introduction

The purpose of this exploratory data analysis is to assess the quality and structure of the delinquency prediction dataset, identify important patterns and anomalies, and highlight variables that may provide early signals of customer delinquency risk. The findings will guide data cleaning and subsequent predictive modeling.

2. Dataset Overview

The dataset contains 500 customer records and 19 variables covering demographic, employment, credit, loan, and payment-history information. The variables include both numerical fields (such as Income, Loan_Balance, Credit_Score, and Credit_Utilization) and categorical fields (such as Employment_Status, Card_Type, and monthly payment-status variables). The dataset contains no duplicate rows or duplicate Customer_ID values.

Key dataset attributes:

- Number of records: 500
- Number of variables: 19
- Key variables: Income, Loan_Balance, Credit_Score, Credit_Utilization, Employment_Status, Card_Type, Month_1–Month_6 payment history, and Delinquent
- Data types: Numerical and categorical
- Target distribution: 16% delinquent and 84% non-delinquent customers

3. Missing Data Analysis

Missing values were identified in Income, Loan_Balance, and Credit_Score. The selected treatment is median imputation because the missing proportions are relatively small and median values are less sensitive to extreme observations.

Variable / Issue	Missing Values	Handling Method & Justification
Income	39 (7.8%)	Median imputation, preferably by employment

		group; preserves records while reducing the influence of extreme incomes.
Loan_Balance	29 (5.8%)	Median imputation after checking the affected records; retains observations without allowing outliers to strongly influence the replacement.
Credit_Score	2 (0.4%)	Median imputation; the missing proportion is very small, making this a simple and low-risk treatment.

4. Key Findings and Risk Indicators

Key findings and risk indicators:

- **Payment history:** Repeated Late or Missed payments are strong behavioral warning signals and may indicate difficulty meeting repayment obligations.
- **Credit utilization:** High utilization may indicate heavy reliance on available credit and reduced financial flexibility.
- **Credit score:** Credit score is a relevant credit-risk measure and should be evaluated as a potential predictor alongside behavioral variables.
- **Loan balance:** A high outstanding loan balance can increase repayment pressure, especially when combined with limited income.
- **Income:** Lower income may indicate reduced capacity to service debt consistently.
- **Employment status:** Unemployment may indicate less stable income and therefore greater potential repayment risk.

Patterns, anomalies, and insights requiring attention:

- Four customers have Credit_Utilization values above 100%; these records should be investigated to determine whether the values are genuine or data-entry errors.
- Employment categories are inconsistent: Employed, employed, and EMP appear to represent the same category and should be standardized before modeling.
- The delinquency target is imbalanced, with only 16% of customers classified as delinquent; model evaluation should therefore consider recall, precision, F1-score, and ROC-AUC rather than accuracy alone.
- Payment behavior across the six monthly history fields may be particularly useful for feature engineering, including total missed payments, total late payments, and recent negative-payment indicators.
- A combination of low income, high loan balance, high credit utilization, and repeated late or missed payments may indicate greater risk than any single factor considered independently.

5. AI & GenAI Usage

Generative AI was used to support the initial exploratory review by summarizing data-quality issues, identifying potential anomalies, and suggesting variables that could be relevant to delinquency prediction. AI-generated observations were treated as exploratory guidance and should be validated during subsequent statistical analysis and model development.

Example AI prompts used:

- “Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency.”
- “Identify the top 3 variables most likely to predict delinquency based on this dataset. Provide brief reasoning.”

6. Conclusion & Next Steps

The EDA identified manageable missing-data issues, inconsistent categorical labels, and several observations requiring anomaly checks. Payment history, credit utilization, credit score, loan balance, income, and employment status are key variables for further investigation as potential delinquency indicators. The next steps are to standardize categorical values, handle missing data, investigate utilization anomalies, engineer payment-history features, and validate the resulting variables using statistical and predictive modeling techniques. Because delinquent customers represent a minority of the dataset, class imbalance should also be considered when training and evaluating models.